

Lab6

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1. Rice theorem: The problem that "any if recursive enumerable language has P property" is undecidable. Property P is non-trivial property.
Non trivial property: P has some (but not all) the description.
Language property $L(m_1) = L(m_2)$, then $\langle m_1 \rangle \in P$ iff $\langle m_2 \rangle \in P$.

2. Assume that P is a decidable language satisfying the properties and let R_P be a TM that decides P .

Then we try to construct TMS that decides A_{TM} using R_P :

① Let M_P be a TM that always rejects, so $L(M_P) = \emptyset$. We can assume that $\langle M_P \rangle \notin P$ without loss of generality because could proceed $\bar{P}$ instead of P if $\langle M_P \rangle \in P$.

② Because P is non-trivial, there exists a TM M_L that accepts L and $L \in P$.

Hence we can assume S to decide A_{TM} to distinguish between M_P and M_L :

$S =$ On input $\langle M, w \rangle$:

1. Use M and w to construct the following TM M_w

$M_w =$ On input x :

1. Simulate M on w . If it halts and rejects, reject.
If it accepts, go to 2.

2. Simulate M_L on x , if it accepts, accept.

2. Use TM R_P to determine whether $\langle M_w \rangle \in P$. If yes, accept; if No, reject.

TM M_w simulates M_L if M accepts w . Hence $L(M_w)$ equals $L(T)$, if M accepts w and $\emptyset$ otherwise. Therefore $\langle M_w \rangle \in P$ iff M accepts w . Thus we get TMS that can decide A_{TM} . However, A_{TM} is undecidable if leads to contradiction, so P is undecidable. Rice theorem is proved.

3. Assume that $\emptyset \notin C$, then we can assume that there is a language $L \in C$ semidecided by M_L .

Given a TM M and input w , and we wish to decide whether M halts on w . Construct a TM $T_{M,w}$, such that the language semidecided by $T_{M,w}$ is either the language L or the language $\emptyset$.

$T_{M,w}$: On input x

1. Simulate the universal TM U on input " M " " w ". If it halts, then goes to step 2.

2. Simulate M_L on x .

Naturally, if $U("M", "w") = \uparrow$, $T_{M,w}(x) = \uparrow$ as well.

$T_{M,w}(x)$: if $U("M", "w") \neq \uparrow$, then $M_L(x)$ else $\uparrow$

The language semidecided by $T_{M,w}$ is in class C iff M halts on w . Suppose that M halts on input w , $T_{M,w}$ on input x determines this, and then always goes to accept x iff $x \in L$. Hence in the case, the language semidecided by $T_{M,w}$ is L which is in C .

Suppose that $M(w) = \uparrow$. In this case, $T_{M,w}$ never halts, and thus M semidecides the language $\emptyset$.

4. An oracle is a TM with access to an oracle, a hypothetical device that can instantly decide whether a given string belongs to a specific language. The oracle answers queries made by the machine during its computation.

A language A is Turing-reducible to language B if there exists a TM with access to an oracle for B that can decide A . This means that solving B allows the machine to also solve A .

5. $E_{TM} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) = \emptyset \}$

$A_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ accepts } w \}$

Construct a TM T which can decide A_{TM}

$T =$ On input $\langle M \rangle$, M is a TM

1. Construct a TM S

$S =$ On any input

1. Run M on all strings in Σ^* .

2. If M accepts, accept.

2. Query the oracle to determine whether $\langle N, 0 \rangle \in A_{TM}$.

3. If oracle answers NO, accept, If YES, reject.

If M 's language is not empty, N will accept every input and input 0. Hence the oracle will answer YES, and T will reject. If M 's language is empty, T will accept. Thus T decides E_{TM} . Since T also decides A_{TM} , $E_{TM} \leq A_{TM}$ is proved.
